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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 30th January Morning Shift

JEE Main Question Paper with Answer Key

30 Jan 2023	90	22	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	17 questions
Physics	29 questions
Out of syllabus	22 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

Open solution

To inhibit the growth of tumours, identify the compounds used from the following :

A. EDTA

B. Coordination Compounds of Pt

C. D - Penicillamine

D. Cis - Platin

Choose the correct answer from the option given below :

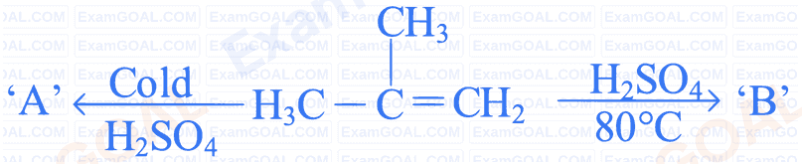
(A) A and B Only **(B)** C and D Only **(C)** B and D Only **(D)** A and C Only

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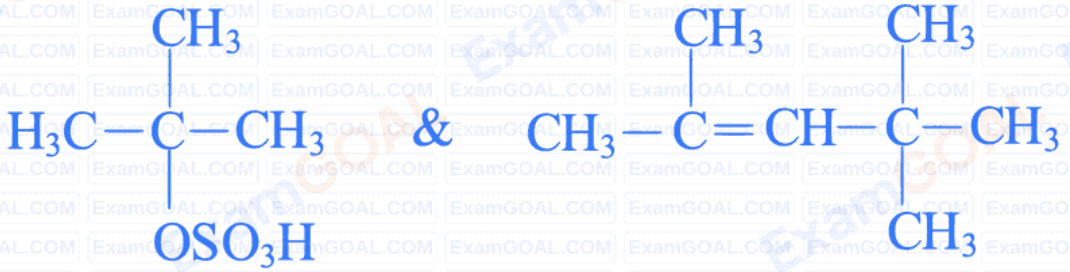
Q2. MCQ (Single Correct Answer)

Open solution

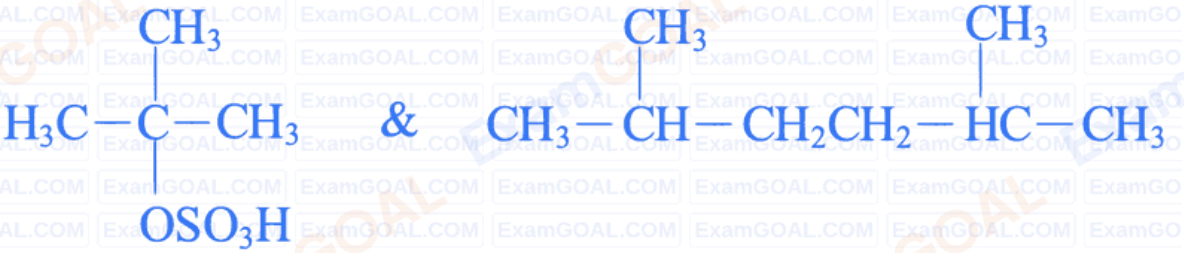
The major products 'A' and 'B', respectively, are



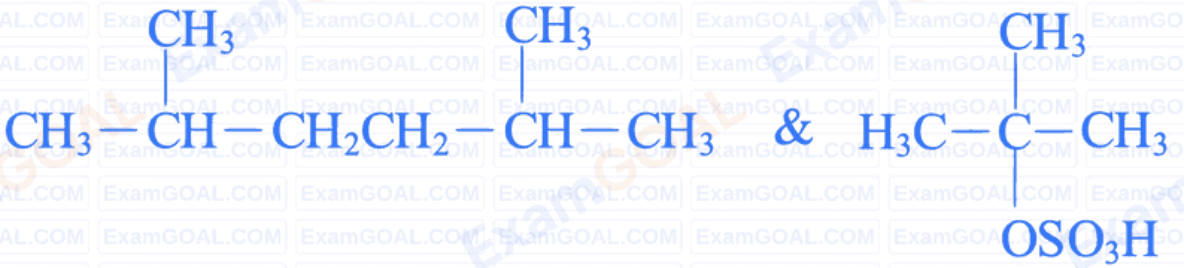
(A)



(B)



(C)



(D)



Q3. MCQ (Single Correct Answer)

Open solution

In the wet tests for identification of various cations by precipitation, which transition element cation doesn't belong to group IV in qualitative inorganic analysis?

(A) Co^{2+} (B) Ni^{2+} (C) Zn^{2+} (D) Fe^{3+}

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Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Ketoses give Seliwanoff's test faster than Aldoses.

Reason (R) : Ketoses undergo β-elimination followed by formation of furfural.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is false but (R) is true

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

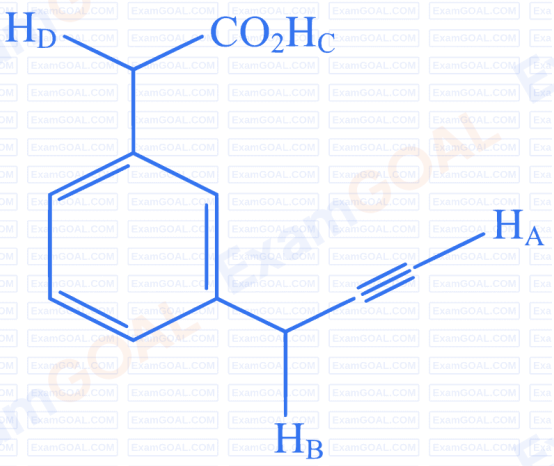
(C) (A) is true but (R) is false

(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

Q5. MCQ (Single Correct Answer)

Open solution

What is the correct order of acidity of the protons marked $\mathrm{A}-\mathrm{D}$ in the given compounds ?



(A) $\mathrm{H_D} > \mathrm{H_C} > \mathrm{H_B} > \mathrm{H_A}$

(B) $\mathrm{H_C} > \mathrm{H_A} > \mathrm{H_D} > \mathrm{H_B}$

(C) $\mathrm{H_C} > \mathrm{H_D} > \mathrm{H_B} > \mathrm{H_A}$

(D) $\mathrm{H_C} > \mathrm{H_D} > \mathrm{H_A} > \mathrm{H_B}$

Q6. MCQ (Single Correct Answer)

Open solution

Which of the following is correct order of ligand field strength?

(A) $\mathrm{S^{2-} < C_2O_4^{2-} < NH_3 < en < CO}$

(B) $\mathrm{CO < en < NH_3 < C_2O_4^{2-} < S^{2-}}$

(C) $\mathrm{S^{2-} < NH_3 < en < CO < C_2O_4^{2-}}$

(D) $\mathrm{NH_3 < en < CO < S^{2-} < C_2O_4^{2-}}$

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Q7. MCQ (Single Correct Answer)

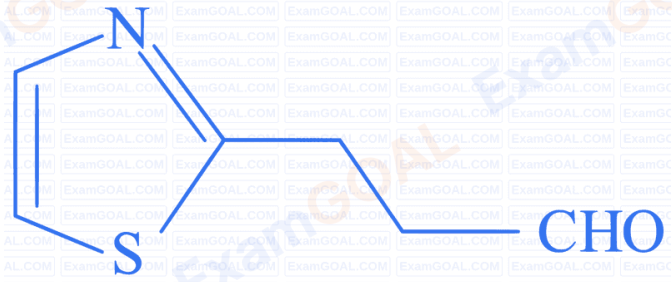
Open solution

Which of the following compounds would give the following set of qualitative analysis ?

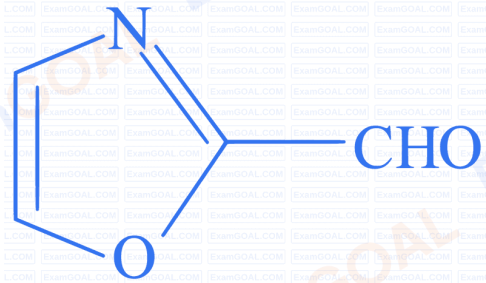
(i) Fehling's Test : Positive

(ii) Na fusion extract upon treatment with sodium nitroprusside gives a blood red colour but not prussian blue.

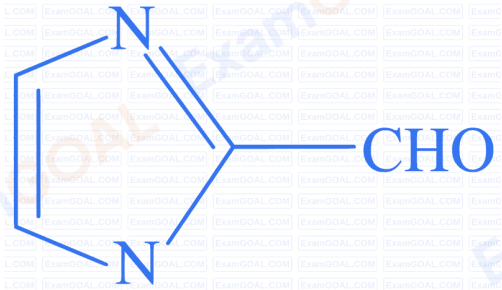
(A)



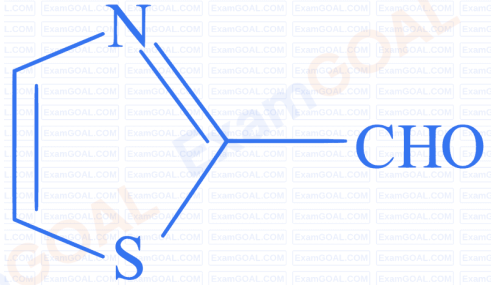
(B)



(C)



(D)



Q8. MCQ (Single Correct Answer)

Open solution

During the qualitative analysis of SO_3^{2-} using dilute H_2SO_4, SO_2 gas is evolved which turns $K_2Cr_2O_7$ solution (acidified with dilute H_2SO_4) :

(A) blue (B) black (C) red (D) green

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Q9. MCQ (Single Correct Answer)

Open solution

For OF_2 molecule consider the following :

A. Number of lone pairs on oxygen is 2 .

B. FOF angle is less than 104.5°.

C. Oxidation state of O is -2.

D. Molecule is bent 'V' shaped.

E. Molecular geometry is linear.

correct options are:

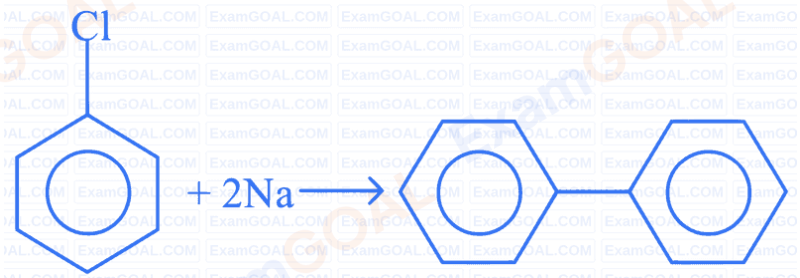
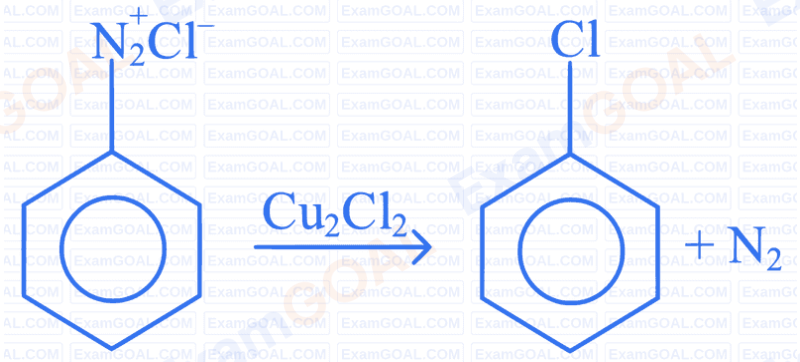
(A) A, C, D only **(B)** A, B, D only **(C)** C, D, E only **(D)** B, E, A only

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Q10. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I		List II
A.		I.	Fittig reaction
B.		II.	Wurtz Fittig reaction
C.		III.	Finkelstein reaction
D.	$C_2H_5Cl + NaI \rightarrow C_2H_5I + NaCl$	IV.	Sandmeyer reaction

Choose the correct answer from the options given below :

(A) A - IV, B - II, C - III, D - I

(B) A - II, B - I, C - IV, D - III

(C) A - II, B - I, C - III, D - IV

(D) A - III, B - II, C - IV, D - I

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Q11. MCQ (Single Correct Answer)

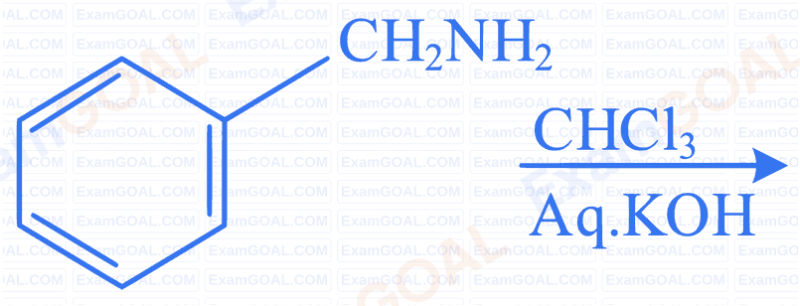
Open solution

Benzyl isocyanide can be obtained by :

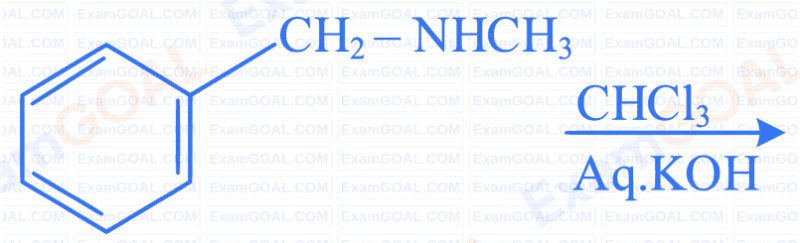
A.



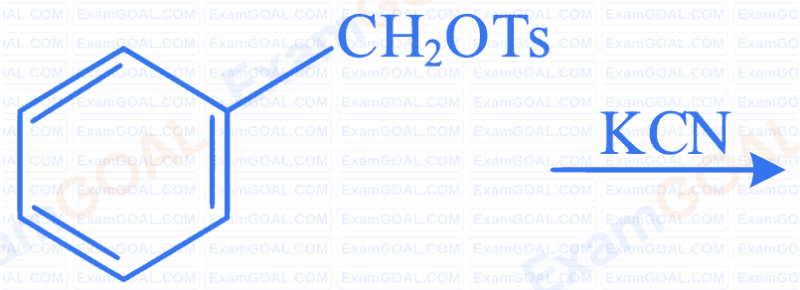
B.



C.



D.



Choose the correct answer from the options given below :

(A) B and C **(B)** A and D **(C)** A and B **(D)** Only B

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Q12. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I (Atomic number)		List II (Block of periodic table)
A.	37	I.	p-block
B.	78	II.	d-block
C.	52	III.	f-block
D.	65	IV.	s-block

Choose the correct answer from the options given below :

(A) A - II, B - IV, C - I, D - III

(B) A - IV, B - III, C - II, D - I

(C) A - IV, B - II, C - I, D - III

(D) A - I, B - III, C - IV, D - II

Q13. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I (molecules/ions)		List II (No. of lone pairs of e^- on central atom)
A.	IF_7	I.	Three
B.	ICl_4^-	II.	One
C.	XeF_6	III.	Two
D.	XeF_2	IV.	Zero

Choose the correct answer from the options given below :

(A) A - II, B - I, C - IV, D - III

(B) A - IV, B - I, C - II, D - III

(C) A - IV, B - III, C - II, D - I

(D) A - II, B - III, C - IV, D - I

Q14. Numerical

Open solution

Some amount of dichloromethane (CH_2Cl_2) is added to $671.141\ mL$ of chloroform $(CHCl_3)$ to prepare $2.6 \times 10^{-3}M$ solution of $CH_2Cl_2(DCM)$. The concentration of DCM is _____ ppm (by mass).

Given :

atomic mass : C = 12

H = 1

Cl = 35.5

density of $CHCl_3 = 1.49\ g\ cm^{-3}$

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Q15. Numerical

If compound A reacts with B following first order kinetics with rate constant $2.011 \times 10^{-3}\ s^{-1}$. The time taken by A (in seconds) to reduce from 7 g to 2 g will be _____. (Nearest Integer)

$[\log 5 = 0.698, \log 7 = 0.845, \log 2 = 0.301]$

Open solution

Q16. Numerical

600 mL of 0.01 M HCl is mixed with 400 mL of 0.01 M H_2SO_4. The pH of the mixture is _____ $\times 10^{-2}$. (Nearest integer)

[Given $\log 2 = 0.30$

$\log 3 = 0.48$

$\log 5 = 0.69$

$\log 7 = 0.84$

$\log 11 = 1.04$]

Open solution

Q17. Numerical

When 2 litre of ideal gas expands isothermally into vacuum to a total volume of 6 litre, the change in internal energy is _____ J. (Nearest integer)

Open solution

Q18. Numerical

Consider the cell

$Pt_{(s)}\ |H_2(g, 1\ atm)|H^+(aq, 1M)||Fe^{3+}(aq), Fe^{2+}(aq)\ |\ Pt(s)$

When the potential of the cell is 0.712 V at 298 K, the ratio $[Fe^{2+}]/[Fe^{3+}]$ is _____. (Nearest integer)

Given : $Fe^{3+} + e^- = Fe^{2+}, E^{\theta}Fe^{3+}, Fe^{2+}\ |\ Pt = 0.771$

$\frac{2.303RT}{F} = 0.06\ V$

Open solution

Q19. Numerical

A trisubstituted compound 'A', $C_{10}H_{12}O_2$ gives neutral $FeCl_3$ test positive. Treatment of compound 'A' with NaOH and CH_3Br gives $C_{11}H_{14}O_2$, with hydroiodic acid gives methyl iodide and with hot conc. NaOH gives a compound B, $C_{10}H_{12}O_2$. Compound 'A' also decolorises alkaline $KMnO_4$. The number of π bond/s present in the compound 'A' is _____.

Open solution

Q20. Numerical

The number of electrons involved in the reduction of permanganate to manganese dioxide in acidic medium is _____.

Open solution

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Q21. Numerical

A solution containing $2\ \mathrm{g}$ of a non-volatile solute in $20\ \mathrm{g}$ of water boils at $373.52\ \mathrm{K}$. The molecular mass of the solute is _____ $\mathrm{g\ mol^{-1}}$. (Nearest integer)

Given, water boils at $373\ \mathrm{K}$, $\mathrm{K_b}$ for water $= 0.52\ \mathrm{K\ kg\ mol^{-1}}$

Open solution

Q22. Numerical

The energy of one mole of photons of radiation of frequency $2 \times 10^{12}\ \mathrm{Hz}$ in $\mathrm{J\ mol^{-1}}$ is _____. (Nearest integer)

[Given : $\mathrm{h} = 6.626 \times 10^{-34}\ \mathrm{Js}$

$\mathrm{N_A} = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$]

Open solution

Mathematics 17 questions

Q23. MCQ (Single Correct Answer)

Let the solution curve $y = y(x)$ of the differential equation

$\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp\left\{\frac{x^3 - \tan^{-1} x^3}{\sqrt{(1+x^6)}}\right\}$ pass through the origin. Then $y(1)$ is equal to :

(A) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$

(B) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

(C) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$

(D) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$

Open solution

Q24. MCQ (Single Correct Answer)

The minimum number of elements that must be added to the relation $\mathrm{R} = \{(\mathrm{a}, \mathrm{b}), (\mathrm{b}, \mathrm{c})\}$ on the set $\{a, b, c\}$ so that it becomes symmetric and transitive is :

(A) 7 (B) 3 (C) 4 (D) 5

Open solution

Q25. MCQ (Single Correct Answer)

Let the system of linear equations

$x + y + kz = 2$

$2x + 3y - z = 1$

$3x + 4y + 2z = k$

have infinitely many solutions. Then the system

$(k+1)x + (2k-1)y = 7$

$(2k+1)x + (k+5)y = 10$

has :

(A) unique solution satisfying $x - y = 1$

(B) infinitely many solutions

(C) no solution

(D) unique solution satisfying $x + y = 1$

Open solution

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Q26. MCQ (Single Correct Answer)

If [t] denotes the greatest integer $\leq$ t, then the value of $\frac{3(e-1)}{e}\int_1^2 x^2 e^{[x]+[x^3]}dx$ is :

(A) $e^8 - e$ (B) $e^7 - 1$ (C) $e^9 - e$ (D) $e^8 - 1$

Open solution

Q27. MCQ (Single Correct Answer)

If the coefficient of x^{15} in the expansion of $\left(ax^3 + \frac{1}{bx^{1/3}}\right)^{15}$ is equal to the coefficient of x^{-15} in the expansion of $\left(ax^{1/3} - \frac{1}{bx^3}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b) :

(A) a = 3b (B) ab = 1 (C) ab = 3 (D) a = b

Open solution

Q28. MCQ (Single Correct Answer)

Suppose $f : \mathbb{R} \to (0, \infty)$ be a differentiable function such that $5f(x + y) = f(x) \cdot f(y), \forall x, y \in \mathbb{R}$. If $f(3) = 320$, then $\sum_{n=0}^{5} f(n)$ is equal to :

(A) 6875 (B) 6525 (C) 6575 (D) 6825

Open solution

Q29. MCQ (Single Correct Answer)

If the solution of the equation $\log_{\cos x} \cot x + 4\log_{\sin x} \tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha+\sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha + \beta$ is equal to :

(A) 3 (B) 6 (C) 4 (D) 5

Open solution

Q30. MCQ (Single Correct Answer)

If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is :

(A) $5 - \frac{3}{2}\sqrt{3}$ (B) $4 - 2\sqrt{3}$ (C) 2 (D) 4

Open solution

Q31. MCQ (Single Correct Answer)

Let a unit vector $\widehat{OP}$ make angles α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in \left(0, \frac{\pi}{2}\right)$. If $\widehat{OP}$ is perpendicular to the plane through points $(1, 2, 3), (2, 3, 4)$ and $(1, 5, 7)$, then which one of the following is true?

(A) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(B) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(C) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(D) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

Open solution

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Q32. MCQ (Single Correct Answer)

If an unbiased die, marked with $-2, -1, 0, 1, 2, 3$ on its faces, is thrown five times, then the probability that the product of the outcomes is positive, is :

(A) $\frac{27}{288}$ (B) $\frac{521}{2592}$ (C) $\frac{440}{2592}$ (D) $\frac{881}{2592}$

Open solution

Q33. Numerical

$\lim\limits_{x\to 0} \frac{48}{x^4} \int\limits_0^x \frac{t^3}{t^6+1}\, dt$ is equal to _____.

Open solution

Q34. Numerical

Number of 4-digit numbers (the repetition of digits is allowed) which are made using the digits 1, 2, 3 and 5, and are divisible by 15, is equal to _____.

Open solution

Q35. Numerical

Let $z = 1 + i$ and $z_1 = \frac{1+i\bar{z}}{\bar{z}(1-z)+\frac{1}{z}}$. Then $\frac{12}{\pi}\arg(z_1)$ is equal to _____.

Open solution

Q36. Numerical

Let α be the area of the larger region bounded by the curve $y^2 = 8x$ and the lines $y = x$ and $x = 2$, which lies in the first quadrant. Then the value of 3α is equal to _____.

Open solution

Q37. Numerical

Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one-one functions $f : \mathrm{S} \to \mathrm{P(S)}$, where $\mathrm{P(S)}$ denote the power set of S, such that $f(n) \subset f(\mathrm{m})$ where $n < m$ is _____.

Open solution

Q38. Numerical

Let $f^1(x) = \frac{3x+2}{2x+3}, x \in \mathbf{R} - \left\{\frac{-3}{2}\right\}$ For $\mathrm{n} \geq 2$, define $f^{\mathrm{n}}(x) = f^1 o f^{\mathrm{n}-1}(x)$. If $f^5(x) = \frac{\mathrm{a}x+\mathrm{b}}{\mathrm{b}x+\mathrm{a}}, \gcd(\mathrm{a}, \mathrm{b}) = 1$, then $\mathrm{a} + \mathrm{b}$ is equal to _____.

Open solution

Q39. Numerical

The mean and variance of 7 observations are 8 and 16 respectively. If one observation 14 is omitted and a and b are respectively mean and variance of remaining 6 observations, then $\mathrm{a} + 3\mathrm{b} - 5$ is equal to _____.

Open solution

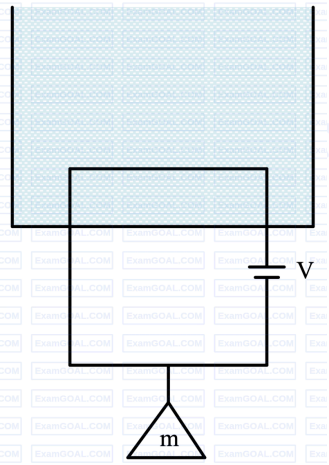
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Physics 29 questions

Q40. MCQ (Single Correct Answer)

A massless square loop, of wire of resistance 10Ω, supporting a mass of $1\ \mathrm{g}$, hangs vertically with one of its sides in a uniform magnetic field of $10^3\mathrm{G}$, directed outwards in the shaded region. A dc voltage V is applied to the loop. For what value of V, the magnetic force will exactly balance the weight of the supporting mass of $1\ \mathrm{g}$?

(If sides of the loop $= 10\ \mathrm{cm},\ \mathrm{g} = 10\ \mathrm{ms}^{-2}$)



Open solution

(A) 1 V (B) $\frac{1}{10}$ V (C) 10 V (D) 100 V

Q41. MCQ (Single Correct Answer)

The height of liquid column raised in a capillary tube of certain radius when dipped in liquid A vertically is, $5\ \mathrm{cm}$. If the tube is dipped in a similar manner in another liquid B of surface tension and density double the values of liquid A, the height of liquid column raised in liquid B would be _____ m.

Open solution

(A) 0.05 (B) 0.20 (C) 0.5 (D) 0.10

Q42. MCQ (Single Correct Answer)

A small object at rest, absorbs a light pulse of power $20\ \mathrm{mW}$ and duration $300\ \mathrm{ns}$. Assuming speed of light as $3 \times 10^8\ \mathrm{m/s}$, the momentum of the object becomes equal to :

Open solution

(A) $1 \times 10^{-17}\ \mathrm{kg\,m/s}$ (B) $0.5 \times 10^{-17}\ \mathrm{kg\,m/s}$

(C) $3 \times 10^{-17}\ \mathrm{kg\,m/s}$ (D) $2 \times 10^{-17}\ \mathrm{kg\,m/s}$

Q43. MCQ (Single Correct Answer)

Choose the correct relationship between Poisson ratio (σ), bulk modulus (K) and modulus of rigidity (η) of a given solid object :

Open solution

(A) $\sigma = \frac{3K+2\eta}{6K+2\eta}$ (B) $\sigma = \frac{3K-2\eta}{6K+2\eta}$

(C) $\sigma = \frac{6K+2\eta}{3K-2\eta}$ (D) $\sigma = \frac{6K-2\eta}{3K-2\eta}$

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Q44. MCQ (Single Correct Answer)

The pressure (P) and temperature (T) relationship of an ideal gas obeys the equation $\mathrm{PT}^2 =$ constant. The volume expansion coefficient of the gas will be :

(A) $3T^2$ (B) $\frac{3}{T^2}$ (C) $\frac{3}{T^3}$ (D) $\frac{3}{T}$

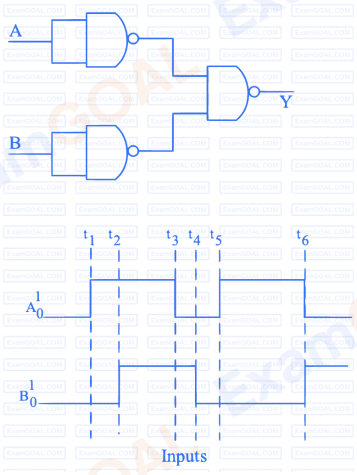
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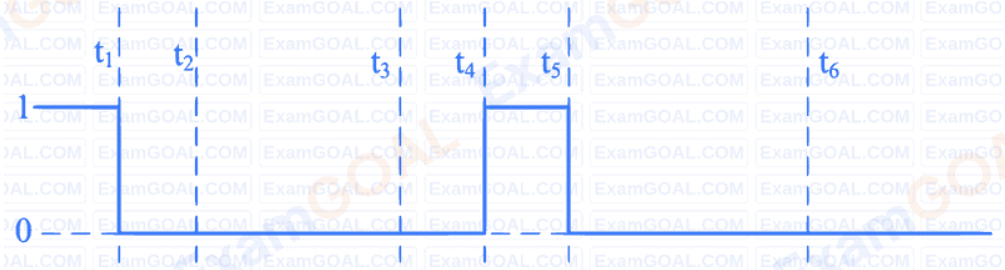
Q45. MCQ (Single Correct Answer)

Open solution

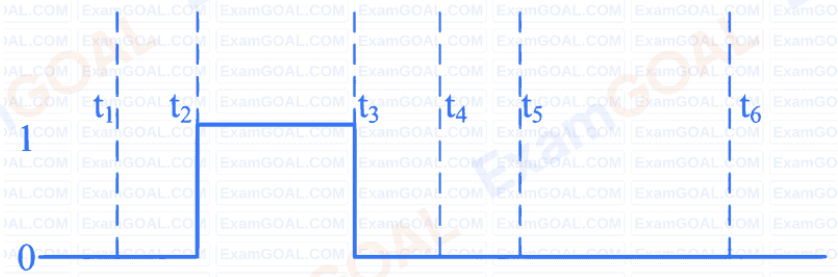
The output waveform of the given logical circuit for the following inputs A and B as shown below, is :



(A)



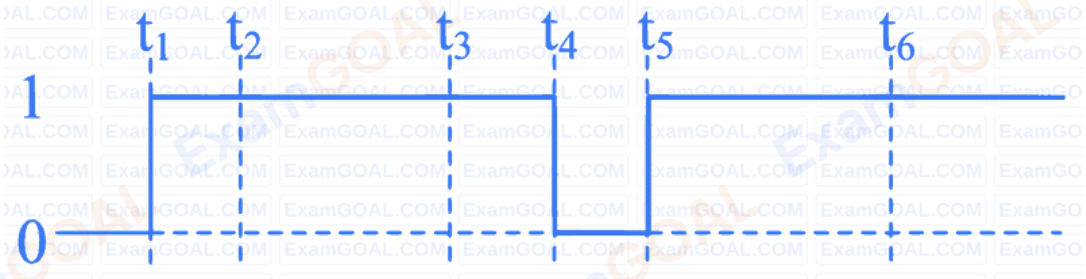
(B)



(C)



(D)



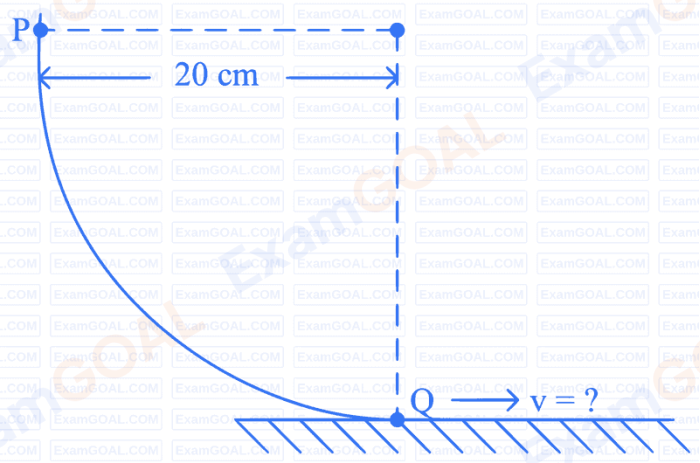
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Q46. MCQ (Single Correct Answer)

Open solution

As per the given figure, a small ball P slides down the quadrant of a circle and hits the other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assume the collision to be elastic, the velocity of ball Q after collision will be :

($g = 10\ m/s^2$)



(A) 0.25 m/s (B) 4 m/s (C) 0 (D) 2 m/s

Q47. MCQ (Single Correct Answer)

Open solution

The charge flowing in a conductor changes with time as $Q(t) = \alpha t - \beta t^2 + \gamma t^3$. Where α, β and γ are constants. Minimum value of current is :

(A) $\beta - \frac{\alpha^2}{3\gamma}$ (B) $\alpha - \frac{3\beta^2}{\gamma}$

(C) $\alpha - \frac{\beta^2}{3\gamma}$ (D) $\alpha - \frac{\gamma^2}{3\beta}$

Q48. MCQ (Single Correct Answer)

Open solution

Heat is given to an ideal gas in an isothermal process.

A. Internal energy of the gas will decrease.

B. Internal energy of the gas will increase.

C. Internal energy of the gas will not change.

D. The gas will do positive work.

E. The gas will do negative work.

Choose the correct answer from the options given below :

(A) B and D only (B) C and E only (C) A and E only (D) C and D only

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Q49. MCQ (Single Correct Answer)

Electric field in a certain region is given by $\overrightarrow{\mathrm{E}}=\left(\frac{\mathrm{A}}{x^2}\hat{i}+\frac{\mathrm{B}}{y^3}\hat{j}\right)$. The SI unit of A and B are :

(A) $\mathrm{Nm^2C};\mathrm{Nm^3C}$

(B) $\mathrm{Nm^3C^{-1}};\mathrm{Nm^2C^{-1}}$

(C) $\mathrm{Nm^3C};\mathrm{Nm^2C}$

(D) $\mathrm{Nm^2C^{-1}};\mathrm{Nm^3C^{-1}}$

Open solution

Q50. MCQ (Single Correct Answer)

Two isolated metallic solid spheres of radii R and $2\mathrm{R}$ are charged such that both have same charge density σ. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ'. The ratio $\frac{\sigma'}{\sigma}$ is :

(A) $\frac{5}{3}$

(B) $\frac{5}{6}$

(C) $\frac{9}{4}$

(D) $\frac{4}{3}$

Open solution

Q51. MCQ (Single Correct Answer)

Speed of an electron in Bohr's 7th orbit for Hydrogen atom is $3.6\times10^6\ \mathrm{m/s}$. The corresponding speed of the electron in 3rd orbit, in $\mathrm{m/s}$ is :

(A) $\left(1.8\times10^6\right)$

(B) $\left(7.5\times10^6\right)$

(C) $\left(8.4\times10^6\right)$

(D) $\left(3.6\times10^6\right)$

Open solution

Q52. MCQ (Single Correct Answer)

The magnetic moments associated with two closely wound circular coils A and B of radius $\mathrm{r_A}=10\ \mathrm{cm}$ and $\mathrm{r_B}=20\ \mathrm{cm}$ respectively are equal if : (Where $\mathrm{N_A},\mathrm{I_A}$ and $\mathrm{N_B},\mathrm{I_B}$ are number of turns and current of A and B respectively)

(A) $4\,\mathrm{N_A I_A}=\mathrm{N_B I_B}$

(B) $2\,\mathrm{N_A I_A}=\mathrm{N_B I_B}$

(C) $\mathrm{N_A}=2\,\mathrm{N_B}$

(D) $\mathrm{N_A I_A}=4\,\mathrm{N_B I_B}$

Open solution

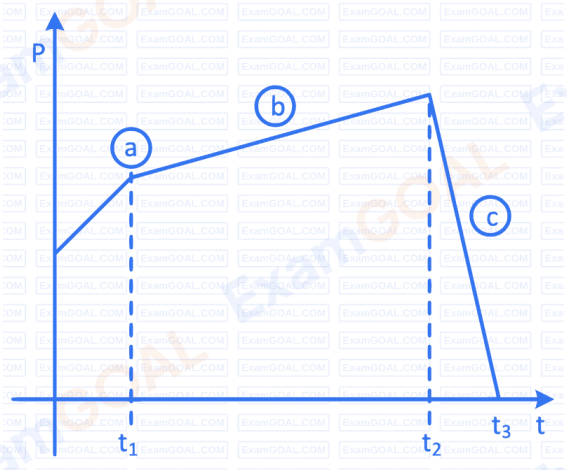
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Q53. MCQ (Single Correct Answer)

Open solution

The figure represents the momentum time $(\mathrm{p}-\mathrm{t})$ curve for a particle moving along an axis under the influence of the force. Identify the regions on the graph where the magnitude of the force is maximum and minimum respectively?

If $(t_3 - t_2) < t_1$



(A) b and c (B) c and a (C) a and b (D) c and b

Q54. MCQ (Single Correct Answer)

Open solution

If the gravitational field in the space is given as $\left(-\frac{K}{r^2}\right)$. Taking the reference point to be at $\mathrm{r} = 2\,\mathrm{cm}$ with gravitational potential $\mathrm{V} = 10\,\mathrm{J/kg}$. Find the gravitational potential at $\mathrm{r} = 3\,\mathrm{cm}$ in SI unit (Given, that $\mathrm{K} = 6\,\mathrm{Jcm/kg}$)

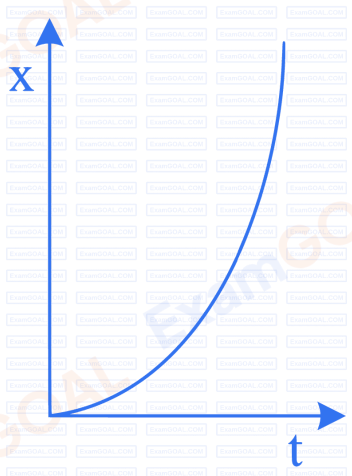
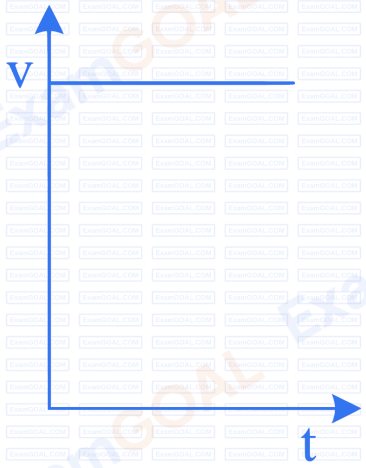
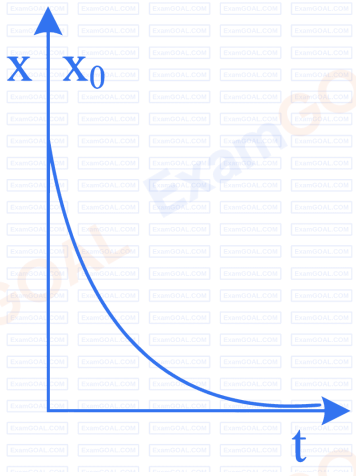
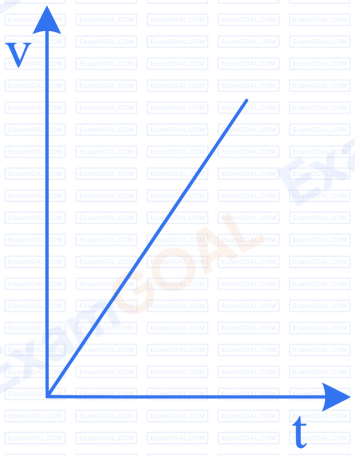
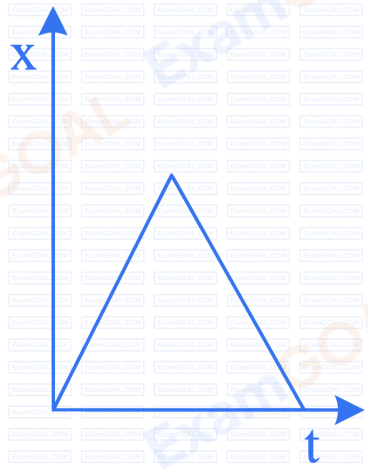
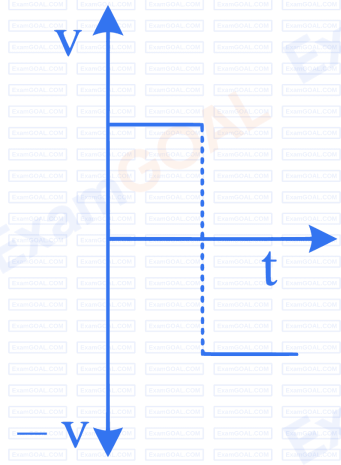
(A) 9 (B) 11 (C) 10 (D) 12

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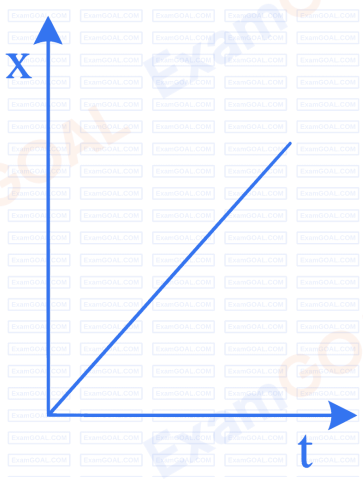
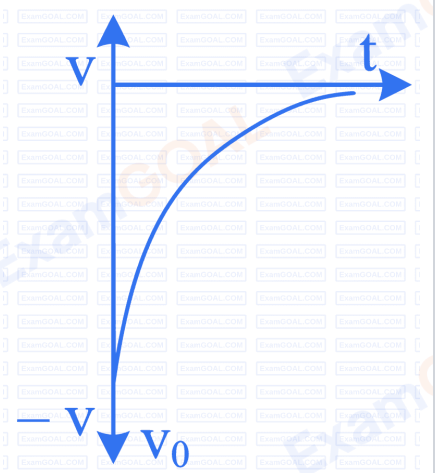
Q55. MCQ (Single Correct Answer)

Open solution

Match Column-I with Column-II :

	Column-I ($_{\mathfrak{X}}$-t graphs)		Column-II ($_{\Omega}$-t graphs)
A.		I.	
B.		II.	
C.		III.	

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	Column-I ($_{\mathfrak{X}}$-t graphs)		Column-II ($_{\Omega}$-t graphs)
D.		IV.	

Choose the correct answer from the options given below:

(A) A- II, B-III, C-IV, D-I

(B) A- II, B-IV, C-III, D-I

(C) A- I, B-III, C-IV, D-II

(D) A- I, B-II, C-III, D-IV

Q56. MCQ (Single Correct Answer)

A ball of mass $200\ \mathrm{g}$ rests on a vertical post of height $20\ \mathrm{m}$. A bullet of mass $10\ \mathrm{g}$, travelling in horizontal direction, hits the centre of the ball. After collision both travels independently. The ball hits the ground at a distance $30\ \mathrm{m}$ and the bullet at a distance of $120\ \mathrm{m}$ from the foot of the post. The value of initial velocity of the bullet will be (if $g = 10\ \mathrm{m/s^2}$) :

Open solution

(A) 120 m/s

(B) 360 m/s

(C) 400 m/s

(D) 60 m/s

Q57. MCQ (Single Correct Answer)

A person has been using spectacles of power -1.0 dioptre for distant vision and a separate reading glass of power 2.0 dioptres. What is the least distance of distinct vision for this person :

Open solution

(A) 50 cm

(B) 40 cm

(C) 30 cm

(D) 10 cm

Q58. MCQ (Single Correct Answer)

In a series LR circuit with $\mathrm{X_L} = \mathrm{R}$, power factor P_1. If a capacitor of capacitance C with $\mathrm{X_C} = \mathrm{X_L}$ is added to the circuit the power factor becomes P_2. The ratio of P_1 to P_2 will be :

Open solution

(A) $1 : \sqrt{2}$

(B) 1 : 3

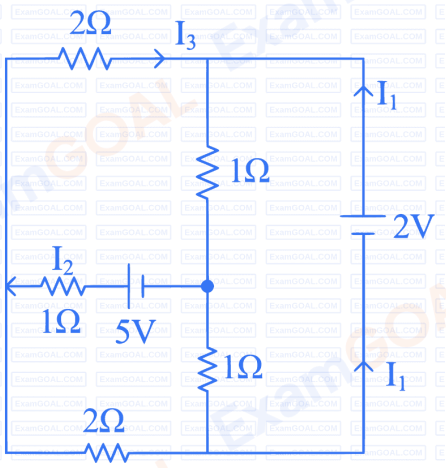
(C) 1 : 2

(D) 1 : 1

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Q59. Numerical

In the following circuit, the magnitude of current I_1, is _____ A.



Open solution

Q60. Numerical

In an experiment for estimating the value of focal length of converging mirror, image of an object placed at $40\ \text{cm}$ from the pole of the mirror is formed at distance $120\ \text{cm}$ from the pole of the mirror. These distances are measured with a modified scale in which there are 20 small divisions in $1\ \text{cm}$. The value of error in measurement of focal length of the mirror is $\frac{1}{\text{K}}\ \text{cm}$. The value of K is _____.

Open solution

Q61. Numerical

The general displacement of a simple harmonic oscillator is $x = A\sin\omega t$. Let T be its time period. The slope of its potential energy (U) - time (t) curve will be maximum when $t = \frac{T}{\beta}$. The value of β is _____.

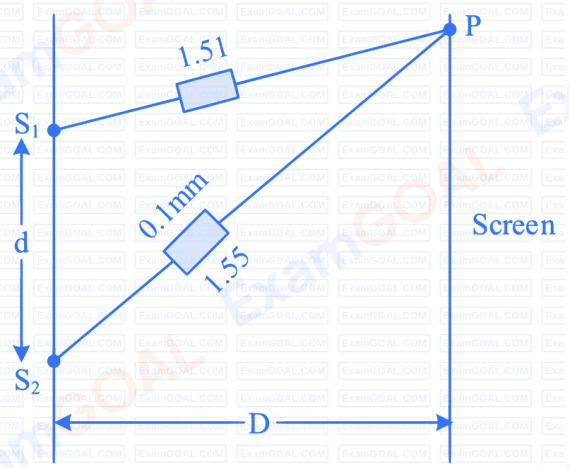
Open solution

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Q62. Numerical

In Young's double slit experiment, two slits S_1 and S_2 are 'd' distance apart and the separation from slits to screen is D (as shown in figure). Now if two transparent slabs of equal thickness $0.1\ \text{mm}$ but refractive index 1.51 and 1.55 are introduced in the path of beam ($\lambda = 4000\ \overset{o}{A}$) from S_1 and S_2 respectively. The central bright fringe spot will shift by _____ number of fringes.

Open solution



Q63. Numerical

A thin uniform rod of length $2\ \text{m}$, cross sectional area 'A' and density 'd' is rotated about an axis passing through the centre and perpendicular to its length with angular velocity ω. If value of ω in terms of its rotational kinetic energy E is $\sqrt{\frac{\alpha E}{Ad}}$ then value of α is _____.

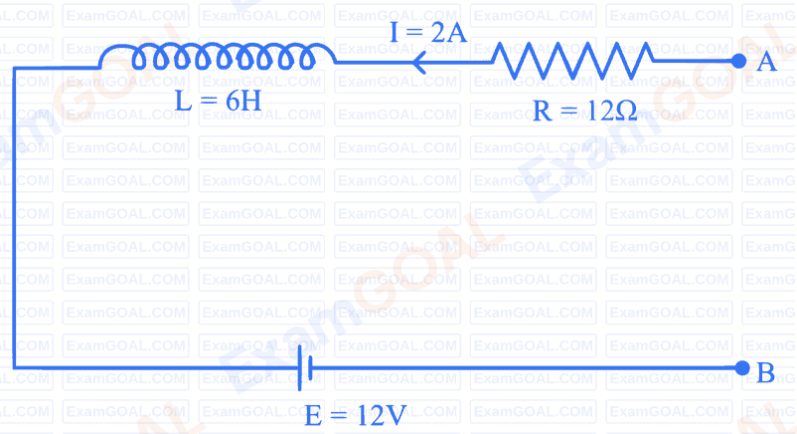
Open solution

Q64. Numerical

A horse rider covers half the distance with $5\ \text{m/s}$ speed. The remaining part of the distance was travelled with speed $10\ \text{m/s}$ for half the time and with speed $15\ \text{m/s}$ for other half of the time. The mean speed of the rider averaged over the whole time of motion is $\frac{x}{7}\ \text{m/s}$. The value of x is _____.

Open solution

Q65. Numerical



As per the given figure, if $\frac{\text{dI}}{\text{dt}} = -1\ \text{A}/s$ then the value of V_{AB} at this instant will be _____ V.

Open solution

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Q66. Numerical

In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by $0.5\ \text{mm}$ on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46th division the circular scale coincide with the reference line. The diameter of the wire is _____ $\times 10^{-2}\ \text{mm}$.

Open solution

Q67. Numerical

A point source of light is placed at the centre of curvature of a hemispherical surface. The source emits a power of $24\ \text{W}$. The radius of curvature of hemisphere is $10\ \text{cm}$ and the inner surface is completely reflecting. The force on the hemisphere due to the light falling on it is _____ $\times 10^{-8}\ \text{N}$.

Open solution

Q68. Numerical

A capacitor of capacitance $900\mu\text{F}$ is charged by a $100\ \text{V}$ battery. The capacitor is disconnected from the battery and connected to another uncharged identical capacitor such that one plate of uncharged capacitor connected to positive plate and another plate of uncharged capacitor connected to negative plate of the charged capacitor. The loss of energy in this process is measured as $x \times 10^{-2}\ \text{J}$. The value of x is _____.

Open solution

Out of syllabus 22 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q69. MCQ (Single Correct Answer) · **Chemistry**

Caprolactam when heated at high temperature in presence of water, gives

(A) Dacron (B) Nylon 6 (C) Teflon (D) Nylon 6,6

Open solution

Q70. MCQ (Single Correct Answer) · **Chemistry**

Formation of photochemical smog involves the following reaction in which A, B and C are respectively.

i. $\text{NO}_2 \xrightarrow{h\nu} \text{A} + \text{B}$
ii. $\text{B} + \text{O}_2 \rightarrow \text{C}$
iii. $\text{A} + \text{C} \rightarrow \text{NO}_2 + \text{O}_2$

Choose the correct answer from the options given below :

(A) NO, O & O_3 (B) N, O_2 & O_3
(C) $\text{O}, \text{N}_2\text{O}$ & NO (D) O, NO & NO_3^-

Open solution

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Q71. MCQ (Single Correct Answer) · **Chemistry**

In the extraction of copper, its sulphide ore is heated in a reverberatory furnace after mixing with silica to:

(A) remove FeO as $FeSiO_3$

(B) remove calcium as $CaSiO_3$

(C) decrease the temperature needed for roasting of Cu_2S

(D) separate CuO as $CuSiO_3$

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

Amongst the following compounds, which one is an antacid?

(A) Ranitidine (B) Brompheniramine (C) Terfenadine (D) Meprobamate

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In expensive scientific instruments, silica gel is kept in watch-glasses or in semipermeable membrane bags.

Reason (R) : Silica gel adsorbs moisture from air via adsorption, thus protects the instrument from water corrosion (rusting) and / or prevents malfunctioning.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) (A) is true but (R) is false

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Lithium aluminium hydride can be prepared from the reaction of :

(A) $LiCl$ and Al_2H_6 (B) LiH and $Al(OH)_3$

(C) LiH and Al_2Cl_6 (D) $LiCl$, Al and H_2

Open solution

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Q75. MCQ (Single Correct Answer) · **Chemistry**

The alkaline earth metal sulphate(s) which are readily soluble in water is/are :

A. $BeSO_4$

B. $MgSO_4$

C. $CaSO_4$

D. $SrSO_4$

E. $BaSO_4$

Choose the correct answer from the options given below :

(A) A and B (B) B and C (C) B only (D) A only

Open solution

Q76. Numerical · **Chemistry**

A $300\ mL$ bottle of soft drink has $0.2\ M\ CO_2$ dissolved in it. Assuming CO_2 behaves as an ideal gas, the volume of the dissolved CO_2 at STP is _____ mL. (Nearest integer)

Given : At STP, molar volume of an ideal gas is $22.7\ L\ mol^{-1}$

Open solution

Q77. MCQ (Single Correct Answer) · **Mathematics**

Among the statements :

$(S1)\ ((p \vee q) \Rightarrow r) \Leftrightarrow (p \Rightarrow r)$

$(S2)\ ((p \vee q) \Rightarrow r) \Leftrightarrow ((p \Rightarrow r) \vee (q \Rightarrow r))$

(A) only (S1) is a tautology (B) neither (S1) nor (S2) is a tautology

(C) both (S1) and (S2) are tautologies (D) only (S2) is a tautology

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

If $a_n = \frac{-2}{4n^2-16n+15}$, then $a_1 + a_2 + \ldots + a_{25}$ is equal to :

(A) $\frac{51}{144}$ (B) $\frac{49}{138}$ (C) $\frac{50}{141}$ (D) $\frac{52}{147}$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive directions of x-axis and y axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y-axis and the area of $\triangle OAB$ is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to :

(A) $\frac{392}{3}$ (B) 98 (C) 196 (D) $\frac{196}{3}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

If $P(h, k)$ be a point on the parabola $x = 4y^2$, which is nearest to the point $Q(0, 33)$, then the distance of P from the directrix of the parabola $y^2 = 4(x + y)$ is equal to :

(A) 8 (B) 2 (C) 6 (D) 4

Open solution

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Q81. MCQ (Single Correct Answer) · **Mathematics**

If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a} = \alpha\vec{b} - \hat{n}, (\alpha \neq 0)$ and $\vec{b}.\vec{c} = 12$, then $\left|\vec{c} \times (\vec{a} \times \vec{b})\right|$ is equal to :

(A) 15 (B) 9 (C) 6 (D) 12

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The coefficient of x^{301} in $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots + x^{500}$ is :

(A) ${}^{500}C_{300}$ (B) ${}^{501}C_{200}$ (C) ${}^{500}C_{301}$ (D) ${}^{501}C_{302}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let $A = \begin{pmatrix} \mathrm{m} & \mathrm{n} \\ \mathrm{p} & \mathrm{q} \end{pmatrix}, \mathrm{d} = |\mathrm{A}| \neq 0$ and $|\mathrm{A} - \mathrm{d}(\mathrm{Adj A})| = 0$. Then

(A) $1 + \mathrm{d}^2 = \mathrm{m}^2 + \mathrm{q}^2$ (B) $1 + d^2 = (m+q)^2$

(C) $(1+d)^2 = m^2 + q^2$ (D) $(1+d)^2 = (m+q)^2$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

The line l_1 passes through the point (2, 6, 2) and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is :

(A) 9 (B) 7 (C) $\frac{19}{3}$ (D) $\frac{13}{3}$

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is :

(A) 2 (B) 3 (C) 4 (D) 0

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let $y = x + 2, 4y = 3x + 6$ and $3y = 4x + 1$ be three tangent lines to the circle $(x-h)^2 + (y-k)^2 = r^2$. Then $h + k$ is equal to :

(A) 6 (B) $5\,(1 + \sqrt{2})$ (C) 5 (D) $5\sqrt{2}$

Open solution

Q87. Numerical · **Mathematics**

Let $\sum_{n=0}^{\infty} \frac{\mathrm{n}^3((2\mathrm{n})!) + (2\mathrm{n}-1)(\mathrm{n}!)}{(\mathrm{n}!)((2\mathrm{n})!)} = \mathrm{ae} + \frac{\mathrm{b}}{\mathrm{e}} + \mathrm{c}$, where $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathbb{Z}$ and $e = \sum_{\mathrm{n}=0}^{\infty} \frac{1}{\mathrm{n}!}$ Then $\mathrm{a}^2 - \mathrm{b} + \mathrm{c}$ is equal to _____.

Open solution

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Q88. Numerical · **Mathematics**

If the equation of the plane passing through the point $(1, 1, 2)$ and perpendicular to the line $x - 3y + 2z - 1 = 0 = 4x - y + z$ is $\mathrm{A}x + \mathrm{B}y + \mathrm{C}z = 1$, then $140(\mathrm{C} - \mathrm{B} + \mathrm{A})$ is equal to _____.

Open solution

Q89. Numerical · **Mathematics**

If $\lambda_1 < \lambda_2$ are two values of λ such that the angle between the planes $P_1 : \vec{r}(3\hat{i} - 5\hat{j} + \hat{k}) = 7$ and $P_2 : \vec{r} \cdot (\lambda\hat{i} + \hat{j} - 3\hat{k}) = 9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1, 10\lambda_2, 2)$ to the plane P_1 is _____.

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

A sinusoidal carrier voltage is amplitude modulated. The resultant amplitude modulated wave has maximum and minimum amplitude of 120 V and 80 V respectively. The amplitude of each sideband is :

(A) 15 V (B) 20 V (C) 5 V (D) 10 V

Open solution

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Answer Key

Chemistry

1. **C**	2. **A**	3. **D**	4. **C**	5. **D**
6. **A**	7. **A**	8. **D**	9. **B**	10. **B**
11. **C**	12. **C**	13. **C**	14. **148**	15. **623**
16. **186**	17. **0**	18. **10**	19. **4**	20. **3**
21. **100**	22. **798**			

Mathematics

23. **B**	24. **A**	25. **D**	26. **A**	27. **B**
28. **D**	29. **C**	30. **D**	31. **A**	32. **B**
33. **12**	34. **21**	35. **9**	36. **22**	37. **3240**
38. **3125**	39. **37**			

Physics

40. **C**	41. **A**	42. **D**	43. **B**	44. **D**
45. **D**	46. **D**	47. **C**	48. **D**	49. **D**
50. **B**	51. **C**	52. **D**	53. **D**	54. **B**
55. **B**	56. **B**	57. **A**	58. **A**	59. **1.5**
60. **32**	61. **8**	62. **10**	63. **3**	64. **50**
65. **30**	66. **220**	67. **4**	68. **225**	

Out of syllabus

69. **B**	70. **A**	71. **A**	72. **A**	73. **A**
74. **C**	75. **A**	76. **1362**	77. **B**	78. **C**
79. **A**	80. **C**	81. **D**	82. **B**	83. **D**
84. **A**	85. **C**	86. **C**	87. **26**	88. **15**
89. **315**	90. **D**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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